24. $\displaystyle\int_{(-2,3,1)}^{(0,0,0)} 2xz\,dx + 2yz\,dy + (x^2 + y^2)\,dz$

In Problems 25 and 26, evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$.

25. $\mathbf{F}(x, y, z) = (y - yz \sin x)\mathbf{i} + (x + z \cos x)\mathbf{j} + y \cos x\mathbf{k}$; $\mathbf{r}(t) = 2t\mathbf{i} + (1 + \cos t)^2\mathbf{j} + 4 \sin^3 t\mathbf{k}$, $0 \le t \le \pi/2$

26. $\mathbf{F}(x, y, z) = (2 - e^z)\mathbf{i} + (2y - 1)\mathbf{j} + (2 - xe^z)\mathbf{k}$; $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}$, $(-1, 1, -1)$ to $(2, 4, 8)$

27. The inverse square law of gravitational attraction between two masses m_1 and m_2 is given by $\mathbf{F} = -Gm_1m_2\mathbf{r}/\|\mathbf{r}\|^3$, where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Show that $\mathbf{F}$ is conservative. Find a potential function for $\mathbf{F}$.

28. Find the work done by the force $\mathbf{F}(x, y, z) = 8xy^3z\mathbf{i} + 12x^2y^2z\mathbf{j} + 4x^2y^3\mathbf{k}$ acting along the helix $\mathbf{r}(t) = 2 \cos t\mathbf{i} + 2 \sin t\mathbf{j} + t\mathbf{k}$ from $(2, 0, 0)$ to $(1, \sqrt{3}, \pi/3)$. From $(2, 0, 0)$ to $(0, 2, \pi/2)$. [*Hint*: Show that $\mathbf{F}$ is conservative.]

29. If $\mathbf{F}$ is a conservative force field, show that the work done along any simple closed path is zero.

30. A particle in the plane is attracted to the origin with a force $\mathbf{F} = \|\mathbf{r}\|^n\mathbf{r}$, where n is a positive integer and $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$ is the position vector of the particle. Show that $\mathbf{F}$ is conservative. Find the work done in moving the particle between (x_1, y_1) and (x_2, y_2).

31. Suppose $\mathbf{F}$ is a conservative force field with potential function ϕ. In physics the function $p = -\phi$ is called **potential energy**. Since $\mathbf{F} = -\nabla p$, Newton's second law becomes

$$m\mathbf{r}'' = -\nabla p \quad \text{or} \quad m\frac{d\mathbf{v}}{dt} + \nabla p = \mathbf{0}.$$

By integrating $m\dfrac{d\mathbf{v}}{dt} \cdot \dfrac{d\mathbf{r}}{dt} + \nabla p \cdot \dfrac{d\mathbf{r}}{dt} = 0$ with respect to t, derive the law of conservation of mechanical energy: $\frac{1}{2}mv^2 + p = \text{constant}$. [*Hint*: See Problem 39 in Exercises 9.8.]

32. Suppose that C is a smooth curve between points A (at $t = a$) and B (at $t = b$) and that p is potential energy, defined in Problem 31. If $\mathbf{F}$ is a conservative force field and $K = \frac{1}{2}mv^2$ is kinetic energy, show that $p(B) + K(B) = p(A) + K(A)$.

9.10 Double Integrals

Introduction In Section 9.8 we gave the five steps leading to the definition of the definite integral $\int_a^b f(x)\,dx$. The analogous steps that lead to the concept of the *two-dimensional definite integral*, known simply as the **double integral** of a function f of two variables, will be given next.

1. Let $z = f(x, y)$ be defined in a closed and bounded region R of 2-space.
2. By means of a grid of vertical and horizontal lines parallel to the coordinate axes, form a partition P of R into n rectangular subregions R_k of areas ΔA_k that lie entirely in R.
3. Let $\|P\|$ be the **norm** of the partition or the length of the longest diagonal of the R_k.
4. Choose a sample point (x_k^*, y_k^*) in each subregion R_k. See **FIGURE 9.10.1**.
5. Form the sum $\displaystyle\sum_{k=1}^{n} f(x_k^*, y_k^*)\,\Delta A_k$.

Thus we have the following definition:

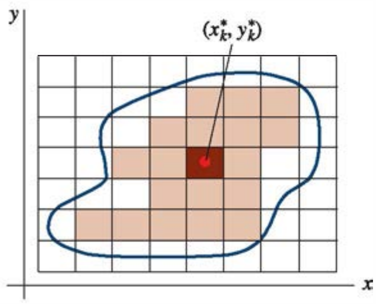


FIGURE 9.10.1 Sample point in kth rectangle

Definition 9.10.1 The Double Integral

Let f be a function of two variables defined on a closed region R of 2-space. Then the **double integral of f over R** is given by

$$\iint_R f(x, y)\,dA = \lim_{\|P\| \to 0} \sum_{k=1}^{n} f(x_k^*, y_k^*)\,\Delta A_k. \tag{1}$$

Integrability If the limit in (1) exists, we say that f is **integrable** over R and that R is the **region of integration**. When f is continuous on R, then f is necessarily integrable over R.

Area When $f(x, y) = 1$ on R, then $\lim_{\|P\| \to 0} \sum_{k=1}^{n} \Delta A_k$ simply gives the **area** A of the region; that is,

$$A = \iint_R dA. \tag{2}$$

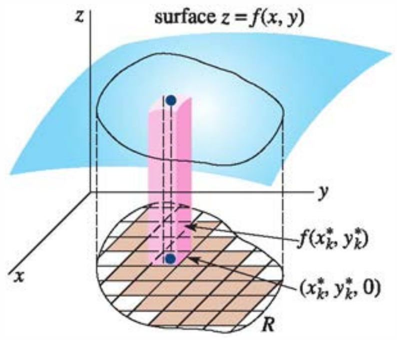


FIGURE 9.10.2 Volume under a surface

Volume

If $f(x, y) \geq 0$ on R, then, as shown in **FIGURE 9.10.2**, the product $f(x_k^*, y_k^*)\,\Delta A_k$ can be interpreted as the volume of a rectangular prism of height $f(x_k^*, y_k^*)$ and base of area ΔA_k. The summation of volumes $\sum_{k=1}^{n} f(x_k^*, y_k^*)\,\Delta A_k$ is an approximation to the **volume** V of the solid *above* the region R and *below* the surface $z = f(x, y)$. The limit of this sum as $\|P\| \to 0$, if it exists, gives the exact volume of this solid; that is, if f is nonnegative on R, then

$$V = \iint_R f(x, y)\, dA. \tag{3}$$

Properties

The double integral possesses the following properties:

Theorem 9.10.1 Properties of Double Integrals

Let f and g be functions of two variables that are integrable over a region R. Then

(*i*) $\displaystyle\iint_R kf(x, y)\, dA = k\iint_R f(x, y)\, dA$, where k is any constant

(*ii*) $\displaystyle\iint_R [f(x, y) \pm g(x, y)]\, dA = \iint_R f(x, y)\, dA \pm \iint_R g(x, y)\, dA$

(*iii*) $\displaystyle\iint_R f(x, y)\, dA = \iint_{R_1} f(x, y)\, dA + \iint_{R_2} f(x, y)\, dA$, where R_1 and R_2 are subregions of R that do not overlap and $R = R_1 \cup R_2$.

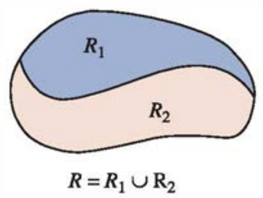


FIGURE 9.10.3 R is the union of two regions

Part (*iii*) of Theorem 9.10.1 is the two-dimensional equivalent of $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$. **FIGURE 9.10.3** illustrates the division of a region into subregions R_1 and R_2 for which $R = R_1 \cup R_2$. R_1 and R_2 can have no points in common except possibly on their common border. Furthermore, Theorem 9.10.1(*iii*) extends to any finite number of nonoverlapping subregions whose union is R.

Regions of Type I and II

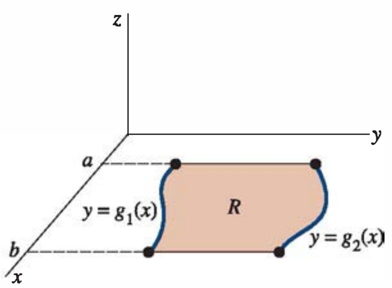
The region shown in **FIGURE 9.10.4(a)**,

$$R: a \leq x \leq b, \quad g_1(x) \leq y \leq g_2(x),$$

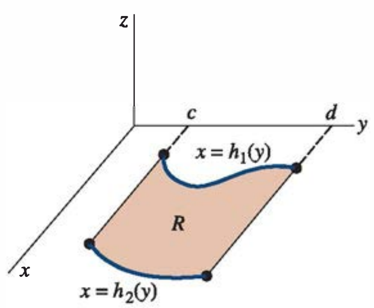
where the boundary functions g_1 and g_2 are continuous, is called a **region of Type I**. In Figure 9.10.4(b), the region

$$R: c \leq y \leq d, \quad h_1(y) \leq x \leq h_2(y),$$

where the functions h_1 and h_2 are continuous, is called a **region of Type II**.



(a) Type I region



(b) Type II region

FIGURE 9.10.4 Regions of integration

Iterated Integrals

Since the partial integral $\int_{g_1(x)}^{g_2(x)} f(x, y)\,dy$ is a function of x alone, we may in turn integrate the resulting function now with respect to x. If f is continuous on a region of Type I, we define an **iterated integral** of f over the region by

$$\int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx = \int_a^b \left[\int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\right] dx. \tag{4}$$

The basic idea in (4) is to carry out *successive integrations*. The partial integral with respect to y gives a function of x, which is then integrated in the usual manner from $x = a$ to $x = b$. The end result of both integrations will be a real number. In a similar manner, we define an iterated integral of a continuous function f over a region of Type II by

$$\int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y)\, dx\, dy = \int_c^d \left[\int_{h_1(y)}^{h_2(y)} f(x, y)\, dx\right] dy. \tag{5}$$

Evaluation of Double Integrals Iterated integrals provide the means for evaluating a double integral $\iint_R f(x, y)\, dA$ over a region of Type I or Type II or a region that can be expressed as a union of a finite number of these regions. The following result is due to the Italian mathematician **Guido Fubini** (1879–1943).

> **Theorem 9.10.2 Fubini's Theorem**
>
> Let f be continuous on a region R.
>
> (*i*) If R is of Type I, then
>
> $$\iint_R f(x, y)\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx. \tag{6}$$
>
> (*ii*) If R is of Type II, then
>
> $$\iint_R f(x, y)\, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y)\, dx\, dy. \tag{7}$$

Theorem 9.10.2 is the double integral analogue of the Fundamental Theorem of Calculus. While Theorem 9.10.2 is difficult to prove, we can get some intuitive feeling for its significance by considering volumes. Let R be a Type I region and $z = f(x, y)$ be continuous and nonnegative on R. The area A of the vertical plane, as shown in **FIGURE 9.10.5**, is the area under the trace of the surface $z = f(x, y)$ in the plane x = constant and hence is given by the partial integral

$$A(x) = \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy.$$

By summing all these areas from $x = a$ to $x = b$, we obtain the volume V of the solid above R and below the surface:

$$V = \int_a^b A(x)\, dx = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx.$$

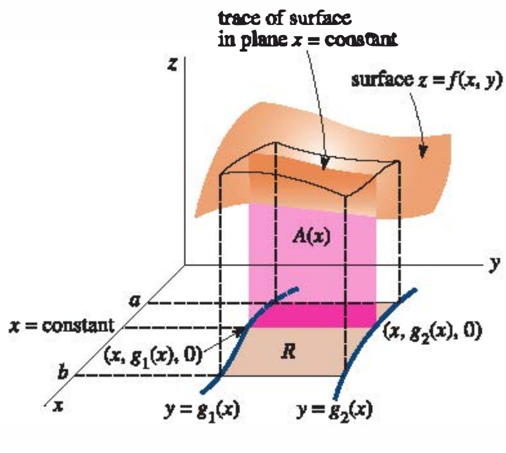


FIGURE 9.10.5 Geometric interpretation of (6)

But, as we have already seen in (3), this volume is also given by the double integral

$$V = \iint_R f(x, y)\, dA.$$

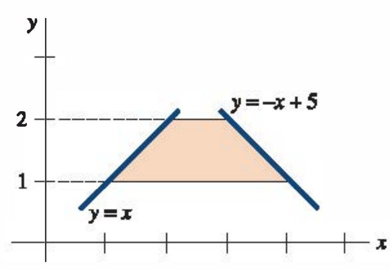


FIGURE 9.10.6 Region of integration in Example 1

EXAMPLE 1 Evaluation of a Double Integral

Evaluate the double integral $\iint_R e^{x+3y}\, dA$ over the region bounded by the graphs of $y = 1, y = 2, y = x$, and $y = -x + 5$. See **FIGURE 9.10.6**.

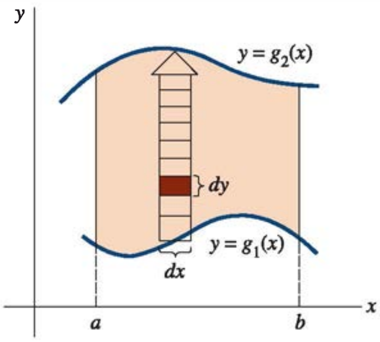
SOLUTION As seen in the figure, the region is of Type II; hence, by (7) we integrate first with respect to x from the left boundary $x = y$ to the right boundary $x = 5 - y$:

$$\iint_R e^{x+3y}\, dA = \int_1^2 \int_y^{5-y} e^{x+3y}\, dx\, dy$$

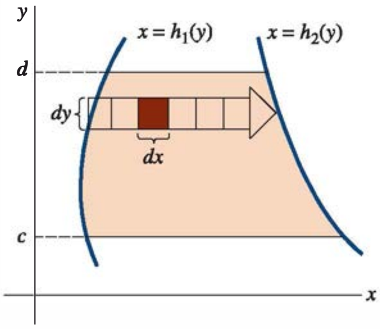
$$= \int_1^2 \Big[e^{x+3y} \Big]_y^{5-y} dy$$

$$= \int_1^2 (e^{5+2y} - e^{4y})\, dy = \left[\frac{1}{2} e^{5+2y} - \frac{1}{4} e^{4y} \right]_1^2$$

$$= \frac{1}{2} e^9 - \frac{1}{4} e^8 - \frac{1}{2} e^7 + \frac{1}{4} e^4 \approx 2771.64. \qquad \equiv$$



(a) Region of Type I



(b) Region of Type II

FIGURE 9.10.7 Summation in y-direction in (a); summation in x-direction in (b)

As an aid in reducing a double integral to an iterated integral with correct limits of integration, it is useful to visualize, as suggested in the foregoing discussion, the double integral as a double summation process. Over a Type I region the iterated integral $\int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx$ is first a summation in the y-direction. Pictorially, this is indicated by the vertical arrow in **FIGURE 9.10.7(a)**; the typical rectangle in the arrow has area $dy\, dx$. The dy placed before the dx signifies that the "volumes" $f(x, y)\, dy\, dx$ of prisms built up on the rectangles are summed vertically with respect to y from the lower boundary curve g_1 to the upper boundary curve g_2. The dx following the dy signifies that the result of each vertical summation is then summed horizontally with respect to x from left ($x = a$) to right ($x = b$). Similar remarks hold for double integrals over regions of Type II. See Figure 9.10.7(b). Recall from (2) that when $f(x, y) = 1$, the double integral $A = \iint_R dA$ gives the area of the region. Thus, Figure 9.10.7(a) shows that $\int_a^b \int_{g_1(x)}^{g_2(x)} dy\, dx$ adds the rectangular areas vertically and then horizontally, whereas Figure 9.10.7(b) shows that $\int_c^d \int_{h_1(y)}^{h_2(y)} dx\, dy$ adds the rectangular areas horizontally and then vertically.

Reversing the Order of Integration

A problem may become easier when the order of integration is **changed** or **reversed**. Also, some iterated integrals that may be impossible to evaluate using one order of integration can perhaps be evaluated using the reverse order of integration.

EXAMPLE 2 Reversing the Order of Integration

Evaluate $\iint_R x e^{y^2}\, dA$ over the region R in the first quadrant bounded by the graphs of $y = x^2$, $x = 0$, $y = 4$.

SOLUTION When the region is viewed as Type I, we have from **FIGURE 9.10.8(a)**, $0 \le x \le 2$, $x^2 \le y \le 4$, and so

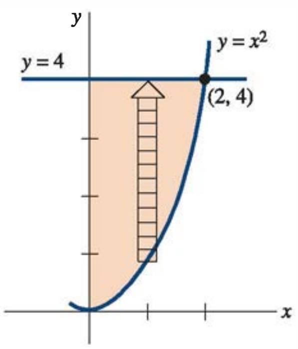
$$\iint_R x e^{y^2} dA = \int_0^2 \int_{x^2}^4 x e^{y^2}\, dy\, dx.$$

The difficulty here is that the partial integral $\int_{x^2}^4 x e^{y^2}\, dy$ cannot be evaluated, since e^{y^2} has no elementary antiderivative with respect to y. However, as we see in Figure 9.10.8(b), we can interpret the same region as a Type II region defined by $0 \le y \le 4, 0 \le x \le \sqrt{y}$. Hence, from (7)

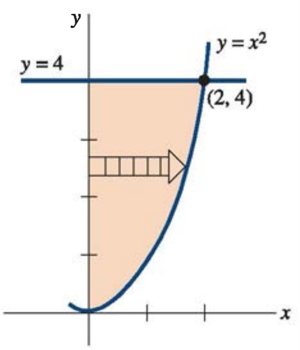
$$\iint_R x e^{y^2} dA = \int_0^4 \int_0^{\sqrt{y}} x e^{y^2}\, dx\, dy$$

$$= \int_0^4 \left[\frac{x^2}{2} e^{y^2} \right]_0^{\sqrt{y}} dy$$

$$= \int_0^4 \frac{1}{2} y e^{y^2} dy = \frac{1}{4} e^{y^2} \Big]_0^4 = \frac{1}{4}(e^{16} - 1). \qquad \equiv$$



(a) Type I region



(b) Type II region

FIGURE 9.10.8 Reversing order of integration in Example 2

□ Laminas with Variable Density—Center of Mass

If ρ is a constant density (mass per unit area), then the mass of the lamina coinciding with a region bounded by the graphs of $y = f(x)$, the x-axis, and the lines $x = a$ and $x = b$ is

$$m = \lim_{\|P\|\to 0} \sum_{k=1}^{n} \rho f(x_k^*)\,\Delta x_k = \int_a^b \rho f(x)\,dx. \tag{8}$$

If a lamina corresponding to a region R has a *variable density* $\rho(x, y)$, where ρ is nonnegative and continuous on R, then analogous to (8) we define its **mass** m by the double integral

$$m = \lim_{\|P\|\to 0} \sum_{k=1}^{n} \rho(x_k^*, y_k^*)\,\Delta A_k = \iint_R \rho(x, y)\,dA. \tag{9}$$

The coordinates of the **center of mass** of the lamina are then

$$\bar{x} = \frac{M_y}{m}, \quad \bar{y} = \frac{M_x}{m}, \tag{10}$$

where

$$M_y = \iint_R x\rho(x, y)\,dA \quad \text{and} \quad M_x = \iint_R y\rho(x, y)\,dA \tag{11}$$

are the **moments** of the lamina about the y- and x-axes, respectively. The center of mass is the point where we consider all the mass of the lamina to be concentrated. If $\rho(x, y)$ is a constant, the center of mass is called the **centroid** of the lamina.

EXAMPLE 3 Center of Mass

A lamina has the shape of the region in the first quadrant that is bounded by the graphs of $y = \sin x$, $y = \cos x$, between $x = 0$ and $x = \pi/4$. Find its center of mass if the density is $\rho(x, y) = y$.

SOLUTION From **FIGURE 9.10.9** we see that

$$m = \iint_R y\,dA = \int_0^{\pi/4}\int_{\sin x}^{\cos x} y\,dy\,dx$$

$$= \int_0^{\pi/4} \frac{y^2}{2}\Bigg]_{\sin x}^{\cos x} dx$$

$$= \frac{1}{2}\int_0^{\pi/4} (\cos^2 x - \sin^2 x)\,dx \quad \leftarrow \text{double angle formula}$$

$$= \frac{1}{2}\int_0^{\pi/4} \cos 2x\,dx = \frac{1}{4}\sin 2x\Bigg]_0^{\pi/4} = \frac{1}{4}.$$

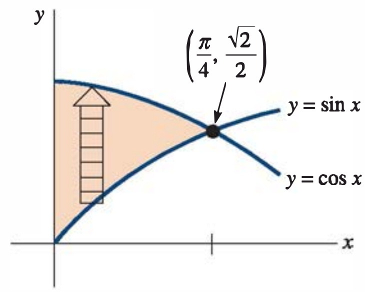


FIGURE 9.10.9 Region in Example 3

Now,

$$M_y = \iint_R xy\,dA = \int_0^{\pi/4}\int_{\sin x}^{\cos x} xy\,dy\,dx$$

$$= \int_0^{\pi/4} \frac{1}{2}xy^2\Bigg]_{\sin x}^{\cos x} dx$$

$$= \frac{1}{2}\int_0^{\pi/4} x\cos 2x\,dx \quad \leftarrow \text{integrate by parts}$$

$$= \left[\frac{1}{4}x\sin 2x + \frac{1}{8}\cos 2x\right]_0^{\pi/4} = \frac{\pi - 2}{16}.$$

Similarly,

$$M_x = \iint_R y^2\, dA = \int_0^{\pi/4}\int_{\sin x}^{\cos x} y^2\, dy\, dx$$

$$= \frac{1}{3}\int_0^{\pi/4} (\cos^3 x - \sin^3 x)\, dx$$

$$= \frac{1}{3}\int_0^{\pi/4} [\cos x(1 - \sin^2 x) - \sin x(1 - \cos^2 x)]\, dx$$

$$= \frac{1}{3}\left[\sin x - \frac{1}{3}\sin^3 x + \cos x - \frac{1}{3}\cos^3 x\right]_0^{\pi/4} = \frac{5\sqrt{2} - 4}{18}.$$

Hence, from (10),

$$\bar{x} = \frac{M_y}{m} = \frac{(\pi - 2)/16}{1/4} \approx 0.29 \quad \text{and} \quad \bar{y} = \frac{M_x}{m} = \frac{(5\sqrt{2} - 4)/18}{1/4} \approx 0.68.$$

Thus the center of mass has the approximate coordinates (0.29, 0.68). ≡

Moments of Inertia The integrals M_x and M_y in (11) are also called the **first moments** of a lamina about the x-axis and y-axis, respectively. The so-called **second moments** of a lamina or **moments of inertia** about the x- and y-axes are, in turn, defined by the double integrals

$$I_x = \iint_R y^2\rho(x, y)\, dA \quad \text{and} \quad I_y = \iint_R x^2\rho(x, y)\, dA. \tag{12}$$

A moment of inertia is the rotational equivalent of mass. For translational motion, kinetic energy is given by $K = \frac{1}{2}mv^2$, where m is mass and v is linear speed. The kinetic energy of a particle of mass m rotating at a distance r from an axis is $K = \frac{1}{2}mv^2 = \frac{1}{2}m(r\omega)^2 = \frac{1}{2}(mr^2)\omega^2 = \frac{1}{2}I\omega^2$, where $I = mr^2$ is its moment of inertia about the axis of rotation and ω is angular speed.

EXAMPLE 4 Moment of Inertia

Find the moment of inertia about the y-axis of the thin homogeneous disk of mass m shown in **FIGURE 9.10.10**.

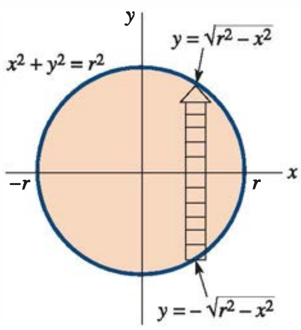


FIGURE 9.10.10 Disk in Example 4

SOLUTION Since the disk is homogeneous, its density is the constant $\rho(x, y) = m/\pi r^2$. Hence, from (12),

$$I_y = \iint_R x^2\left(\frac{m}{\pi r^2}\right) dA = \frac{m}{\pi r^2}\int_{-r}^{r}\int_{-\sqrt{r^2-x^2}}^{\sqrt{r^2-x^2}} x^2\, dy\, dx$$

$$= \frac{2m}{\pi r^2}\int_{-r}^{r} x^2\sqrt{r^2 - x^2}\, dx \quad \leftarrow \text{trigonometric substitution } x = r\sin\theta$$

$$= \frac{2mr^2}{\pi}\int_{-\pi/2}^{\pi/2} \sin^2\theta\cos^2\theta\, d\theta$$

$$= \frac{mr^2}{2\pi}\int_{-\pi/2}^{\pi/2} \sin^2 2\theta\, d\theta$$

$$= \frac{mr^2}{4\pi}\int_{-\pi/2}^{\pi/2} (1 - \cos 4\theta)\, d\theta = \frac{1}{4}mr^2.$$

≡

Radius of Gyration The **radius of gyration** of a lamina of mass m and the moment of inertia I about an axis is defined by

$$R_g = \sqrt{\frac{I}{m}}. \tag{13}$$

Since (13) implies that $I = mR_g^2$, the radius of gyration is interpreted as the radial distance the lamina, considered as a point mass, can rotate about the axis without changing the rotational inertia of the body. In Example 4 the radius of gyration is $R_g = \sqrt{I_y/m} = \sqrt{(mr^2/4)/m} = r/2$.

9.10 Exercises

Answers to selected odd-numbered problems begin on page ANS-23.

In Problems 1–8, evaluate the given partial integral.

1. $\int_{-1}^{3} (6xy - 5e^y)\,dx$ **2.** $\int_{1}^{2} \tan xy\,dy$

3. $\int_{1}^{3x} x^3e^{xy}\,dy$ **4.** $\int_{\sqrt{y}}^{y^3} (8x^3y - 4xy^2)\,dx$

5. $\int_{0}^{2x} \frac{xy}{x^2+y^2}\,dy$ **6.** $\int_{x^3}^{x} e^{2y/x}\,dy$

7. $\int_{\tan y}^{\sec y} (2x + \cos y)\,dx$ **8.** $\int_{\sqrt{y}}^{1} y\ln x\,dx$

In Problems 9–12, sketch the region of integration for the given iterated integral.

9. $\int_0^2\int_1^{2x+1} f(x, y)\,dy\,dx$ **10.** $\int_1^4\int_{-\sqrt{y}}^{\sqrt{y}} f(x, y)\,dx\,dy$

11. $\int_{-1}^3\int_0^{\sqrt{16-y^2}} f(x, y)\,dx\,dy$ **12.** $\int_{-1}^2\int_{-x^2}^{x^2+1} f(x, y)\,dy\,dx$

In Problems 13–22, evaluate the double integral over the region R that is bounded by the graphs of the given equations. Choose the most convenient order of integration.

13. $\iint_R x^3y^2\,dA;\ y = x,\ y = 0,\ x = 1$

14. $\iint_R (x + 1)\,dA;\ y = x,\ x + y = 4,\ x = 0$

15. $\iint_R (2x + 4y + 1)\,dA;\ y = x^2,\ y = x^3$

16. $\iint_R xe^y\,dA;$ R the same as in Problem 13

17. $\iint_R 2xy\,dA;\ y = x^3,\ y = 8,\ x = 0$

18. $\iint_R \frac{x}{\sqrt{y}}\,dA;\ y = x^2 + 1,\ y = 3 - x^2$

19. $\iint_R \frac{y}{1+xy}\,dA;\ y = 0,\ y = 1,\ x = 0,\ x = 1$

20. $\iint_R \sin\frac{\pi x}{y}\,dA;\ x = y^2,\ x = 0,\ y = 1,\ y = 2$

21. $\iint_R \sqrt{x^2+1}\,dA;\ x = y,\ x = -y,\ x = \sqrt{3}$

22. $\iint_R x\,dA;\ y = \tan^{-1}x,\ y = 0,\ x = 1$

23. Consider the solid bounded by the graphs of $x^2 + y^2 = 4$, $z = 4 - y$, and $z = 0$ shown in **FIGURE 9.10.11**. Choose and evaluate the correct integral representing the volume V of the solid.

(a) $4\int_0^2\int_0^{\sqrt{4-x^2}} (4 - y)\,dy\,dx$

(b) $2\int_{-2}^2\int_0^{\sqrt{4-x^2}} (4 - y)\,dy\,dx$

(c) $2\int_{-2}^2\int_0^{\sqrt{4-y^2}} (4 - y)\,dx\,dy$

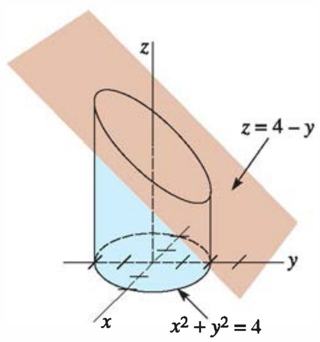


FIGURE 9.10.11 Solid for Problem 23

24. Consider the solid bounded by the graphs of $x^2 + y^2 = 4$ and $y^2 + z^2 = 4$. An eighth of the solid is shown in **FIGURE 9.10.12**. Choose and evaluate the correct integral representing the volume V of the solid.

(a) $4\int_{-2}^2\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (4 - y^2)^{1/2}\,dy\,dx$

(b) $8\int_0^2\int_0^{\sqrt{4-y^2}} (4 - y^2)^{1/2}\,dx\,dy$

(c) $8\int_0^2\int_0^{\sqrt{4-x^2}} (4 - x^2)^{1/2}\,dy\,dx$